

AI-Powered Restaurant Financial Management SaaS

Final Year Major Project + Startup Prototype

- Domain: Restaurant Operations and Finance
- Approach: MERN + AI + SaaS Multi-Tenant Design
- Goal: Real-time financial visibility and smarter decision-making

Problem Statement

- Fragmented systems: POS, spreadsheets, and manual inventory
- No single source of financial truth
- Manual errors and delayed reporting
- Weak cost control and stock planning

Proposed Solution

- Unified dashboard for sales, expense, and inventory
- Automated financial reports and KPIs
- Role-based access for Admin, Manager, Cashier
- AI assistant for forecasting and insights

Architecture

Frontend

React/Next.js dashboard with charts and workflows

Backend

Node.js + Express REST APIs

Database

MongoDB for transactions and tenant data

AI Layer

OCR + LLM + prediction services

Core Modules

- Orders & POS billing
- Expense management with receipt upload
- Inventory tracking and low-stock alerts
- Reports: revenue, profit, top dishes
- AI chat for business Q&A

AI Features

- Smart expense categorization from receipts
- Sales prediction for demand planning
- Margin and anomaly insights
- Natural-language analytics assistant

10-Week Development Plan

- W1-W2: Requirements and UI design
- W3-W6: Backend + auth + core modules
- W7-W8: Frontend integration + AI
- W9-W10: Testing, deployment, documentation

Business and Academic Value

- Strong final-year demo with real-world use case
- Demonstrates full-stack and AI capability
- MVP-ready architecture for startup scaling
- Future scope: subscription billing and multi-branch analytics